

MOBILITY

PROBLEM STATEMENT ID	PROBLEM STATEMENT	PROBLEM STATEMENT DESCRIPTION
MOB-01	Predictive Battery State-of-Health System for Indian EV Two-Wheelers	Develop an AI-based Battery State of Health (SoH) and Remaining Useful Life (RUL) Prediction System that uses NASA/CALCE battery datasets to train an LSTM/Gaussian Process model, applies Indian 2-wheeler duty cycle corrections (city traffic, monsoon charging, heat stress), and provides an interactive fleet dashboard with SoH/RUL predictions, configurable alerts, and model performance metrics (MAE/RMSE).
MOB-02	Reinforcement Learning-Based Adaptive Traffic Signal Control for Indian Intersections	Develop an AI-based Adaptive Traffic Signal Control System using SUMO and DQN/Q-learning to optimize signal timings at a 4-way Indian mixed-traffic intersection based on real-time queue lengths. The system compares performance with fixed-timing and Webster's method under peak and off-peak traffic, evaluates wait time, throughput, and emissions, and optionally uses YOLO-based vehicle counting from real traffic video.

MOB-03	Accessibility-First Multi-Modal Journey Planner Using Indian GTFS Data	Develop an accessible multi-modal journey planner that uses GTFS data (BMTC/BEST/DTC) and OpenStreetMap to provide cost-optimized, accessibility-aware routes with elevators, ramps, and accessible stops. The app features a screen-reader (ARIA) compatible interface, voice guidance, real-time disruption alerts, and supports multi-modal routing using Dijkstra/A* algorithms.
MOB-04	Vehicle-to-Grid (V2G) Energy Management System Simulator for Indian EV Fleets	Develop an AI-based Vehicle-to-Grid (V2G) Energy Management System that simulates an EV fleet and a local grid with solar generation and demand variability using POSOCO data. The system uses LP or RL optimization to schedule charging/discharging cycles, minimizing electricity costs while maintaining vehicle availability, and provides a dashboard comparing cost savings and grid stress reduction with and without V2G.
MOB-05	Autonomous Last-Mile Delivery Robot Simulation for Indian Urban Environments	Develop a ROS2-based Autonomous Delivery Robot Simulation that navigates a simulated Indian urban environment using LiDAR, camera, and SLAM (GMapping/Cartographer) for mapping and localization. The system performs dynamic obstacle avoidance, multi-stop route optimization, and monsoon visibility simulation, while

		evaluating delivery success rate, travel time, and collision metrics in Gazebo/AirSim.
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